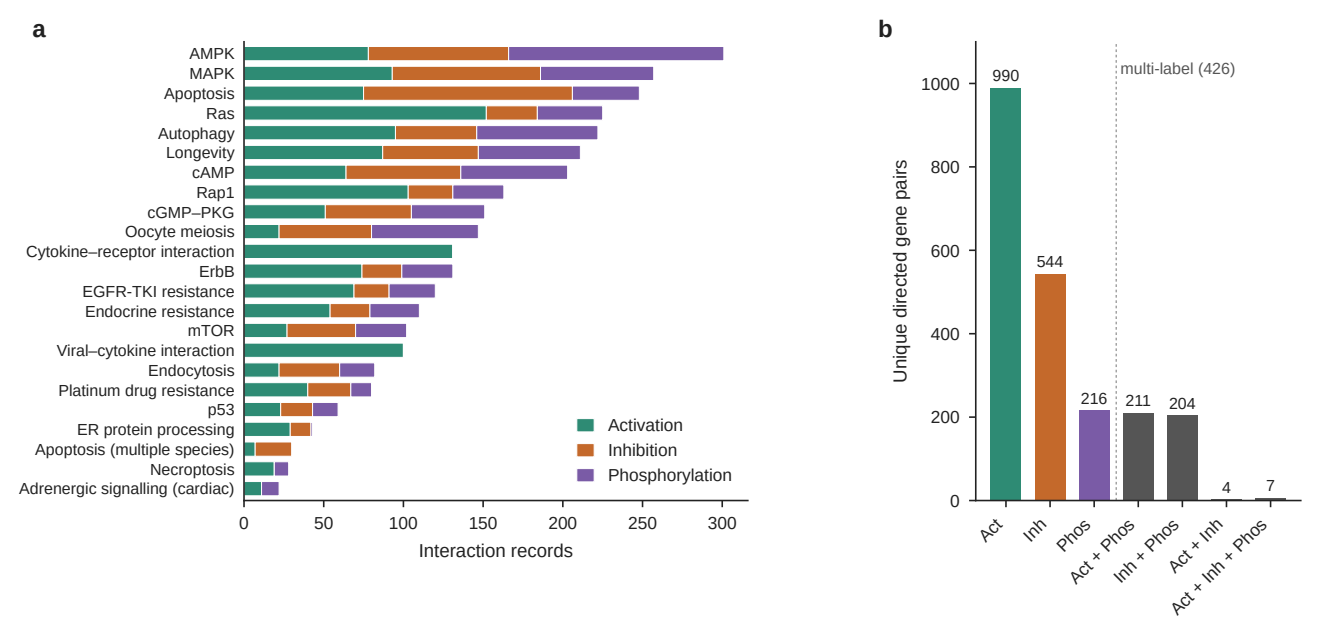
a
AMPK
MAPK
Apoptosis
Ras
Autophagy
Longevity
cAMP
Rap1
cGMP–PKG
Oocyte meiosis
Cytokine–receptor interaction
ErbB
EGFR-TKI resistance
Endocrine resistance
mTOR
Viral–cytokine interaction
Endocytosis
Platinum drug resistance
p53
ER protein processing
Apoptosis (multiple species)
Necroptosis
Adrenergic signalling (cardiac)
0
50
100
150
200
250
300
Interaction records
Activation
Inhibition
Phosphorylation
b
Unique directed gene pairs
0
200
400
600
800
1000
multi-label (426)
990
544
216
211
204
4
7
Act
Inh
Phos
Act + Phos
Inh + Phos
Act + Inh
Act + Inh + Phos